

Gregory Cooke

ASTROPHYSICS · POSTDOCTORAL RESEARCHER

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Employment

Institute of Astronomy, University of Cambridge

Cambridge, UK

Research Associate - Hycean Worlds. PI: Nikku Madhusudhan. 2nd advisor: Emily Mitchell

October 2024 - present

- I am simulating the atmosphere and oceanic environments of Hycean World exoplanets.
- The major component of this work includes the development of new ecological models, inspired by Earth, for exoplanets
- I have supervised three MPhil students and demonstrated independence by guiding them towards producing publishable work with two of the students.

Research Associate in Exoplanetary Atmospheres. PI: Nikku Madhusudhan

August 2023 - September 2024

- My work centred on simulations of exoplanets in the sub-Neptune regime using one-dimensional (1D) and 3D photochemical models.
- I explored the chemical nature of habitable exoplanet atmospheres, assuming various planetary conditions.

Education

University of Leeds

Leeds, UK

PhD in Astrophysics; Thesis title: 3D simulations of oxygenated rocky planetary climates and observational predictions. Advisors: Professor Dan Marsh, Dr Catherine Walsh.

October 2019 - July 2023

- My thesis focused on simulating rocky worlds and understanding their climates, chemistry, and habitability. I used and modified the Community Earth System Model ([CESM2](#)), mostly the Whole Atmosphere Community Climate Model ([WACCM6](#)) configuration.
- I simulated early Earth with a younger Sun and with varied atmospheric oxygen (O₂) concentrations.
- I used the Planetary Spectrum Generator ([PSG](#)) to determine how detectable specific planetary properties (e.g. chemical species such as ozone and oxygen; temporal variability) are using the next generation of telescopes (e.g. [LUVOIR](#)).
- I performed simulations for tidally locked M dwarf exoplanets and predicted observations of these exoplanets.
- I found, for the first time, that it is possible for lethal surface concentrations of O₃ to build up on the surface of habitable zone exoplanets.
- My thesis received recognition for Research Excellence from the Dean of Postgraduate Research Studies.
- I was selected competitively as a [Priestley Climate Scholar](#).

University of Manchester

Manchester, UK

MPhys in Physics (First-Class Honours: 81.4%)

October 2015 - June 2019

- Two MPhys projects on habitability metrics for known exoplanets and how to design a mission to optimise searching for life.
- Most optional courses taken were related to astrophysics (e.g. Astrophysical plasmas, General relativity, Exoplanets).

Publications

Published

- Pica-Ciamarra L., Madhusudhan N., **Cooke G. J.**, *et al.*, 2026, A Systematic Search for Trace Molecules in Exoplanet K2-18 b. [10.3847/2041-8213/ae5dcc](#).
- **Cooke G. J.**, Madhusudhan N., & Mitchell E. G., Ecological modelling of hycean worlds, MNRAS, [10.1093/mnras/stag495](#).
- Sainsbury-Martinez F., **Cooke G. J.**, Walsh C., The Response of Planetary Atmospheres to the Impact of Icy Comets III: Atmospheric Escape, submission planned for ApJ. [10.3847/1538-4357/ae5062/meta](#).
- **Cooke G. J.** *et al.*, The Oxygen Valve on Hydrogen Escape Since the Great Oxidation Event, 2026, [10.5194/egusphere-2025-1133](#)
- Madhusudhan N., Constantinou S., Holmberg M. & **Cooke G. J.**, Exploring the Sub-Neptune Frontier with JWST, 2025, [10.1073/pnas.2416194122](#).
- Sainsbury-Martinez F., Walsh C., & **Cooke G. J.**, 2025, The Response of Planetary Atmospheres to the Impact of Icy Comets I: Tidally-Locked exo-Earths, [10.3847/1538-4357/ad96ad/meta](#)
- Bhongade A., Marsh D. R., Sainsbury-Martinez F., & **Cooke G. J.**, 2024, Asymmetries in the simulated ozone distribution on TRAPPIST-1e due to orography. [10.3847/1538-4357/ad8f2f](#)
- **Cooke G. J.** & Madhusudhan N., Considerations for Photochemical Modeling of Possible Hycean Worlds, 2024, [10.3847/1538-4357/ad8cda](#)
- Sainsbury-Martinez F., Walsh C., **Cooke G. J.**, & Marsh D. R., How Land-Mass Distribution Influences The Atmospheric Dynamics of Tidally Locked Terrestrial Exoplanets. [10.3847/1538-4357/ad6d5b](#)
- **Cooke G. J.**, Marsh D. R., Walsh C., & Sainsbury-Martinez F., 2024, Lethal surface ozone concentrations are possible on habitable zone exoplanets, The Planetary Science Journal, [10.3847/PSJ/ad53c3](#).
- Liu B., Marsh D. R., Walsh C., **Cooke G. J.**, & Sainsbury-Martinez F., July 2024, Eccentric Orbits Enhance the Habitability of Earth-like Exoplanets, Monthly Notices of the Royal Astronomical Society, [10.1093/mnras/stae1758](#)
- **Cooke G. J.**, Marsh D. R., Walsh C., & Youngblood A., 2023, Degenerate interpretations of O₃ spectral features in exoplanet atmosphere observations due to stellar UV uncertainties: a 3D case study with TRAPPIST-1e, The Astrophysical Journal, [10.3847/1538-4357/ad0381](#).
- Liu B., Marsh D. R., Walsh C., & **Cooke G. J.**, June 2023, Higher Water Loss on Earth-like Exoplanets in Eccentric Orbits, Monthly Notices of the Royal Astronomical Society, pp. 1491–1502, [10.1093/mnras/stad1828](#).
- Ji A., Kasting J. F., **Cooke G. J.**, *et al.*, Comparison between ozone column depths & methane lifetimes computed by one- & three-dimensional models at different atmospheric O₂ levels. R. Soc. open sci. 10: 230056. [10.1098/rsos.230056](#).
- **Cooke G. J.**, Marsh D. R., Walsh C., Rugheimer S., & Villanueva G. L., January 2023, Variability due to climate & chemistry in observations of oxygenated Earth-analogue exoplanets, Monthly Notices of the Royal Astronomical Society, 518(1), pp. 206–219, [10.1093/mnras/stac2604](#).
- **Cooke G. J.**, Marsh D. R., Walsh C., Black B., & Lamarque J-F. 2022 A revised lower estimate of ozone columns during Earth's oxygenated history. R. Soc. Open Sci. 9: 211165. [10.1098/rsos.211165](#).

Articles in review

JULY 23, 2026

- **Cooke G. J. et al.** The Representation of Women in Science in UK Science Specifications, submission planned for British Journal of Education.
- Kumar A. & **Cooke G. J.**, Simulations of the Evolving Ozone Layer: Implications for the Early Earth and Exoplanets, submitted to Royal Society Open Science.
- Sarkar S., Madhusudhan N., Holmberg M., Rigby F., Moses J., & **Cooke G. J.**, 2025, Title Embargoed, ApJ Letters.

Articles in prep

- **Cooke G. J.** & Madhusudhan N., et al. Title to be confirmed, submission planned for ApJ.
- **Cooke G. J.** & Madhusudhan N., & Mitchell. Title to be confirmed, submission planned for MNRAS.

Awards

University of Cambridge

Cambridge, UK

Research culture nominee

May 2026

- University of Cambridge Research Culture Nominee for exemplary contributions to the EDI committee

EDI departmental award

June 2026

- The Institute of Astronomy Community Service award, for exemplary contributions to the EDI committee

University of Leeds

Leeds, UK

Research excellence in doctoral thesis

October 2019 - April 2023

- Award to reflect outstanding achievement in the thesis and oral examination at a level significantly above what would be expected for the award of a doctoral degree, including major contributions to the field through interdisciplinary work.

Funding

STFC

Cambridge, UK

DiRAC computer time

Awarded April 2026

- 1.95 M CPUh on DiAL3 to use ROCKE-3D and WACCM6 to simulate early Earth climates under different conditions.
- This computer time is for simulations of Earth's paleoclimates which should lead to between two and three papers.

Royal Astronomical Society

Cambridge, UK

Education & Outreach Small Grants Scheme

Awarded October 2025

- £5,000 for the [IncludeHer](#) UK project. This is work analyzing the UK curriculum with the aim to improve scientific teaching - a paper has been produced and is in review.

University of Cambridge

Cambridge, UK

LCLU grant

October 2024 - September 2026

- A 2-year postdoctoral position with extra funding for a year if progress is made.

University of Cambridge

Cambridge, UK

ERC grant

August 2023 - September 2024

- A 3-year postdoctoral position.

University of Leeds

Leeds, UK

STFC studentship

October 2019 - April 2023

- A 3.5-year STFC studentship (approximately worth £75,000).
- Funding for travel and funding for the conference fee to attend the *3rd Eddy Cross Disciplinary Symposium: Sun, Earth, Planet, Space, Atmosphere and Ocean*, in Vail, Colorado, USA (total \$2,800).

Contributed talks

Apr 2026	Cambridge Centre for Climate Science Symposium 2026 , <i>The Oxygen Valve on Hydrogen Escape Since the Great Oxidation Event</i>	Cambridge, UK
Mar 2026	LCLU Science Day 2026 , <i>Ecological modelling of hycean worlds</i>	Cambridge, UK
Mar 2026	RASReach 2026 , <i>Who Gets Taught? Gender Representation in UK Science Curricula</i>	Manchester, UK
Mar 2025	UK Exoplanet Meeting , <i>A chemistry intercomparison for Proxima Centauri b atmospheric simulations.</i>	Leeds, UK
Mar 2024	LCLU Annual Science Day , <i>Lethal surface ozone concentrations are possible on habitable zone exoplanets.</i>	Cambridge, UK
Jan 2024	Rocky Worlds III , <i>Lethal surface ozone concentrations are possible on habitable zone exoplanets.</i>	Zurich, Switzerland
Nov 2023	Habitable Worlds Observatory – UK community workshop , <i>3D simulations of exoplanet climates and observational predictions</i>	Leicester, UK
Jul 2022	Rocky Worlds II , <i>A revised lower estimate of ozone columns during Earth's oxygenated history.</i>	Oxford, UK
Jul 2022	ResCompLeedsCon2022 , <i>Simulations of tidally locked exoplanet atmospheres in 3D.</i>	Leeds, UK
Jun 2022	3rd Eddy Cross Disciplinary Symposium , <i>3D whole-atmosphere modelling of rocky exoplanet systems and synthetic telescope observations.</i>	CO, USA
Jun 2021	CESM Workshop , <i>Viewing the Earth and its exoplanet analogues through time.</i>	Virtual
Apr 2021	UK Exoplanet Meeting , <i>Oxygen's 2.4 billion year control on Earth's atmosphere with consequences for exoplanet biosignatures.</i>	Virtual

Invited and internal seminars

May 2026	Invited, University of Exeter , <i>Ecological modelling of hycean worlds.</i>	Exeter, UK
Apr 2025	Invited, MPIA Heidelberg , <i>Considerations for Photochemical Modeling of Possible Hycean Worlds.</i>	Virtual
Nov 2024	Invited, University of Leeds , <i>Considerations for Photochemical Modeling of Possible Hycean Worlds.</i>	Leeds, UK
Mar 2024	Invited, University of Oxford , <i>Ozone in habitable zone exoplanet atmospheres: observational ambiguities and lethality to life.</i>	Oxford, UK
Mar 2024	Invited, University of Leeds , <i>Ozone in habitable zone exoplanet atmospheres: observational ambiguities and lethality to life.</i>	Leeds, UK
Nov 2023	Internal, University of Cambridge , <i>Imposter syndrome.</i>	Cambridge, UK

Oct 2023	Internal, University of Cambridge , 3D simulations of oxygenated rocky exoplanet atmospheres and observational predictions.	Cambridge, UK
Feb 2023	Invited, University of Edinburgh , A revised lower estimate of ozone columns during Earth's oxygenated history.	Edinburgh, UK
Oct 2022	Internal, University of Leeds , Variability due to climate and chemistry in observations of oxygenated Earth-analogue exoplanets.	Leeds, UK
May 2022	Invited, National Center for Atmospheric Research , A revised lower estimate of ozone columns during Earth's oxygenated history.	CO, USA
Mar 2022	Internal, University of Leeds , A revised lower estimate of ozone columns during Earth's oxygenated history.	Leeds, UK
May 2021	Invited, University of Cambridge , Oxygen's 2.4 billion year control on Earth's atmosphere with consequences for exoplanet biosignatures.	Virtual
Oct 2020	Invited, National Center for Atmospheric Research , Oxygen as a control over 2.4 billion years of atmospheric evolution.	Virtual

Posters

July 2026	Exoplanets VI , The development of the ozone layer under an evolving Sun. Poster PDF here .	Porto, Portugal
Mar 2026	UK Exoplanet Meeting , Ecological modelling of hycean worlds. Poster PDF here .	Bristol, UK
Jan 2026	Rocky Worlds IV , Ecological modelling of ocean worlds. Poster PDF here .	Gronningen, NED
Jul 2025	Exoclimates VII , Hydrocarbons in sub-Neptune atmospheres. Poster PDF here .	Montréal, CAN
Sep 2024	Origins Federation Conference 2024 , Is K2-18 b a mini-Neptune or a Hycean World? Poster PDF here .	Cambridge, UK
Jun 2024	Exoplanets V , Oxygen's control over hydrogen escape on Earth-like exoplanets. Poster PDF here .	Lieden, NL
Jun 2024	Exoplanets V , Is K2-18 b a mini-Neptune or a Hycean World? Poster PDF here	Leiden, NL
Jun 2023	Exoclimates VI , Characterising stellar UV to improve the interpretation of observations: a 3D case study with TRAPPIST-1 e. Poster PDF here .	Exeter, UK
Sep 2022	UK Exoplanet Meeting , Accurate UV stellar spectra measurements required to use O ₃ as an indicator for O ₂ abundance. Virtual poster .	Edinburgh, UK
May 2022	Exoplanets IV , Variability due to climate in observations of oxygenated Earth-analogue exoplanets.	LV, NV, USA
Jun-Jul 2021	European Astronomical Society Annual Meeting , Oxygen's 2.4 billion year control on Earth's atmosphere with consequences for exoplanet biosignatures. Poster PDF here	Virtual
Jun 2021	The Coupling, Energetics, and Dynamics of Atmospheric Regions workshop , Atmospheric escape on oxygenated Earth-like exoplanet atmospheres.	Virtual
Jul 2020	Exoplanets III , Variable detectability of biosignatures on inhabited worlds.	Virtual

Software experience

- I am an advanced user of Python for atmospheric data analysis, e.g., matplotlib, pandas, numpy, and xarray.
- I have used and modified the FORTRAN codes [Atmos](#) and [Photochem](#) which model planetary atmospheres in 1D.
- I have used and developed an open-source 3D climate model model ([CESM2-WACCM6](#)). I have read Fortran-90 code to understand how certain calculations in [WACCM6](#) are made. I modified the Fortran-90 code to set up different planetary conditions (e.g. altered upper boundary conditions, tidally locked the model, and implemented absorption in the Schumann–Runge bands for H₂O and CO₂).
- I used and developed a pipeline to convert [WACCM6](#) output to interact with the Planetary Spectrum Generator ([PSG](#)). I used new methods (where I swapped particular atmospheric components) to analyse the results for the [WACCM6](#) oxygenated scenarios.
- Coding experience in C++ during my master's degree. The final project was to design a chess game using C++.

Teaching

University of Cambridge

Cambridge, UK

MPhil and masters project supervision

October 2024 - present

- I have supervised Akhil Kumar on two projects related to his LCLU Life in the Universe MPhil for which we are writing up two publications.
- I have supervised two other students on the LCLU MPhil and a masters student on projects relating to astrobiology and photochemical modelling. I am working with Alicia Ericsson on an astrobiology paper.

Lectures

August 2025 - January 2026

- "The search for life in the universe" to be delivered at Jesus College and "The atmospheres and remote sensing of exoplanets" delivered to the Institute of Astronomy Summer Intern students

Part III Masters project

October 2024 - June 2025

- Supervised a fourth year (part III) student. The project was on simulating several different hycean candidates assuming they are either uninhabited or inhabited.

Supervisor Stellar Dynamics and Structure of Galaxies

October 2024 - present

- Supervised the third year (Part II) Stellar Dynamics and Structure of Galaxies course delivered by Vasily Belokurov.
- Supervised two student groups of two students.

Supervisor Stars and Stellar Evolution

October 2023 - January 2024

- Supervised the third year (part II) Stars and Stellar Evolution lecture course delivered by Max Pettini.
- Supervised student groups between the sizes of 1-3 students.

University of Leeds

Leeds, UK

Introductory python course

September 2022

- Introduction to Python lesson during a Community Earth System Model (CESM) tutorial.
- I demonstrated data visualisation using Xarray, Matplotlib, and Cartopy in functions combined with IPyWidgets in a Jupyter notebook.

Lab demonstrating

October 2019 - May 2022

- I taught experiments in the Phys 10001 undergraduate laboratory to 1st year students including: the determination of Planck's constant; measurement of Earth's magnetic field, spectrometer measurement of sodium lines; the viscosity of glycerine; and electrical circuits.
- I marked lab workbooks and formal reports on several of these experiments.

Informal MPhys student supervision

October 2021 - March 2022

- I aided B. Butcher to produce and analyse transmission spectra of Jupiter-sized exoplanets.
- I helped I. Willis analyse WACCM data and produce figures using Python.

Organisation and citizenship

Academic journal reviews

2023 - present

- Nature
- Monthly Notices of the Royal Astronomical Society
- Scientific reports
- The Astrophysical Journal
- Communications earth & environment
- JGR atmospheres

Institute of Astronomy, University of Cambridge

Cambridge, UK

Women's Network co-chair

January 2025 - present

- International Women's Day (IWD) Outreach event
- Organised IWD departmental event
- Organised monthly meetings for the women's network

LCLU Science Day Scientific Organising Committee

March 2026

- As part of a team of four we assessed the submitted abstracts to decide on which ones should be awarded talks or posters.
- The day involved hosting the conference and chairing a specific session on planets.

Work-life balance EDI chair

October 2024 - present

- I arranged and chaired the work-life balance equality diversity and inclusivity (EDI) group.
- I arranged meetings and social events to help improve the work-life balance of students and postdoctoral researchers at the Institute of Astronomy.
- Attended institute meetings on EDI in order to improve the ethos of the workplace.

First year PhD journal club

October 2024 - May 2025

- I organized the journal club for the first year PhD cohort of 12 students.
- I chaired the meetings and facilitated discussion on recent and important papers in different astronomical fields.

University of Leeds

Leeds, UK

Internal seminars chair

January 2020 - October 2022

- I arranged and chaired internal seminars for the University of Leeds Astrophysics group.
- I organised and led weekly informal science sessions where members of the group get together to discuss their current work.
- I led a journal club that ran every three weeks.

University of Leeds Priestley scholars

January 2020 - December 2021

- I attended multiple seminars on interdisciplinary topics relating to climate change, including transport, climate finance, climate modelling, and climate justice.
- I co-organised a seminar on climate finance, as well as a monthly journal club focussed on climate science topics.

University of Manchester Men's Hockey Club

Manchester, UK

Treasurer

May 2017 - May 2018

- I was elected out from a club of approximately 80 members.
- I managed ~£20,000 in financial transactions between the club, club members, the Athletic Union, and several different organisations.

Public engagement and press

- Imperial College London, *Who Gets Taught? Gender Representation in UK Science Curricula*, (May 2026).
- Pint of Science, *The search for life beyond our solar system*, (May 2026).
- International Women's Day outreach activities for year 8 students (March 2026).
- International Women's Day outreach activities for year 8 students (March 2025).
- Invited talk at Bradford Astronomical Society (November 2024).
- Public talk on *Jill Tarter and the Search for ExtraTerrestrial Intelligence* (March 2024).
- Public talk on *A Brief History of Women in Astronomy* for International Women's Day (March 2024).
- Invited talk at Sidney Sussex Wilson-Walker Natural Sciences Society (March 2024).
- Invited talk at Harrogate Astronomical Society (February 2024).
- Public talk at the Institute of Astronomy, University of Cambridge, *Exoplanet Atmospheres* (November 2023). [YouTube](#).
- Invited talk at Wakefield and District Astronomical Society (July 2023).
- Invited talk at Bradford Astronomical Society (April 2023).
- [Everything Astronomy](#) virtual session for Xavier Space Solutions (February 2022).
- I have written a number of astronomy news articles for the astronomy magazine [Popular Astronomy](#).

- TikTok Video summarizing my research for COP 26 and how it is important for understanding our planet (2021).
- [Live YouTube talk](#) for the University of Leeds Be Curious festival on planet habitability (2021).
- [Priestley Scholar Twitter spotlight](#). I was retweeted by the Priestley Scholar Twitter account for a whole day as I tweeted about my research and scientific interests (2021).